

JH Data Project

May 9, 2026

1 Analyzing Death Rate Disparities by Race, Sex, and Time in New York City

Overview: For this project, I will be exploring the most common causes of deaths in New York City, focusing on how they vary by sex and race over the period from 2007 to 2021. The dataset, New York City Leading Causes of Death, can be found on the NYC Open Data website.

I will be answering the following 3 questions in my analysis:

1. Which race/ethnicity groups have the highest age-adjusted death rates for the top leading causes of death in New York City?
2. Within the most affected race/ethnicity groups, how do death rates differ between males and females?
3. How have disparities in age-adjusted death rates between race/ethnicity groups changed over time for the top causes of death?

Dataset: The dataset consists of 7 columns, including the year of death (ranging from 2007 to 2021), leading cause of death, sex, race/ethnicity, number of deaths, death rate, and age-adjusted death rate.

Note: Age-adjusted death rates are used to fairly compare groups by removing the effects of age.

Below are the import packages I will be utilizing to clean, transform, manipulate, and complete the analysis:

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
In [2]: df = pd.read_csv('New_York_City_Leading_Causes_of_Death_20260420.csv')
```

Initial Data Inspection:

Upon previewing the dataset using `.head()` (shown below), you can see that the data is a bit messy. There are inconsistencies in the Sex column where values like “Male” and “M” or “Female” and “F” are mixed. There are also periods used as values, duplicates, extra whitespace, and other non-standard formatting issues.

In the Leading Cause column, I’m planning to remove the ICD (International Classification of Diseases) codes to make the data cleaner and improve grouping and visualization since I’m not doing anything with that information in my analysis.

```
In [3]: print(df.head(-25))
```

	Year	Leading Cause	Sex \
0	2021	Mental and Behavioral Disorders due to Acciden...	Male
1	2021	All Other Causes	Female
2	2021	Chronic Liver Diseases and Cirrhosis	Male
3	2021	Diseases of Heart (I00-I09, I11, I13, I20-I51)	Female
4	2021	Influenza (Flu) and Pneumonia (J09-J18)	Male
...	...	...	...
2072	2007	Chronic Liver Disease and Cirrhosis (K70, K73)	M
2073	2007	Aortic Aneurysm and Dissection (I71)	F
2074	2007	All Other Causes	F
2075	2007	Diabetes Mellitus (E10-E14)	M
2076	2007	Alzheimer's Disease (G30)	F

	Race Ethnicity	Deaths	Death Rate	Age Adjusted Death Rate
0	Other Race/ Ethnicity	9	NaN	NaN
1	Not Stated/Unknown	102	NaN	NaN
2	Not Stated/Unknown	16	NaN	NaN
3	Hispanic	1409	111	95.5
4	Not Stated/Unknown	16	NaN	NaN
...	...	...	...	...
2072	Asian and Pacific Islander	18	3.7	3.6
2073	Not Stated/Unknown	.	.	.
2074	Not Stated/Unknown	27	.	.
2075	Not Stated/Unknown	9	.	.
2076	Not Stated/Unknown	.	.	.

```
[2077 rows x 7 columns]
```

In addition, the data types in the columns Deaths, Death Rate, and Age Adjusted Death Rate will have to be converted. I will be converting the 3 columns values into a numeric type (like int or float).

```
In [4]: print(df.dtypes)
```

```
Year                int64
Leading Cause       object
Sex                object
Race Ethnicity     object
Deaths             object
Death Rate         object
Age Adjusted Death Rate object
dtype: object
```

```
In [5]: print(df.columns)
```

```
Index(['Year', 'Leading Cause', 'Sex', 'Race Ethnicity', 'Deaths',
      'Death Rate', 'Age Adjusted Death Rate'],
      dtype='object')
```

2 Data Cleaning

```
In [6]: df_copy = df.copy()
```

```
In [7]: ### Rename Race Ethnicity column to Race/Ethnicity
```

```
df_copy = df_copy.rename(columns = {'Race Ethnicity': 'Race/Ethnicity'})
```

```
In [8]: ### Keep values in the Sex column consistent with the mapping
```

```
df_copy['Sex'] = df_copy['Sex'].replace({'M': 'Male', 'F': 'Female'})
```

```
In [9]: ### Replace '.' value with NaN
```

```
df_copy = df_copy.replace('.', np.nan)
```

```
In [10]: ### Remove ICD Codes with str.replace function and Regex for removing any text in par
```

```
df_copy['Leading Cause'] = df_copy['Leading Cause'].str.replace(r"\s*\(.*\) ", "", regex=Fa
```

```
In [11]: ### Remove white space in the Race/Ethnicity Column values
```

```
df_copy['Race/Ethnicity'] = df_copy['Race/Ethnicity'].str.replace('/ ', '/', regex=Fa
```

```
In [12]: ### There are quite a few duplicate and lengthy texts in the Race/Ethnicity Column
```

```
sorted(df['Race Ethnicity'].unique())
```

```
Out[12]: ['Asian and Pacific Islander',
          'Black Non-Hispanic',
          'Hispanic',
          'Non-Hispanic Black',
          'Non-Hispanic White',
          'Not Stated/Unknown',
          'Other Race/ Ethnicity',
          'White Non-Hispanic']
```

```
In [13]: ### Change duplicate and lengthy values in the Race/Ethnicity Column to have consiste
```

```
clean_col = {
    'Black Non-Hispanic': 'Non-Hispanic Black',
    'Non-Hispanic Black': 'Non-Hispanic Black',
    'White Non-Hispanic': 'Non-Hispanic White',
```

```

        'Non-Hispanic White': 'Non-Hispanic White',

        'Asian and Pacific Islander': 'Asian/Pacific Islander',

        'Other Race/Ethnicity': 'Other',
        'Not Stated/Unknown': 'Unknown'
    }

    df_copy['Race/Ethnicity'] = df_copy['Race/Ethnicity'].replace(clean_col)

```

In [14]: *### Result of cleaning the above column*

```
sorted(df_copy['Race/Ethnicity'].unique())
```

```

Out[14]: ['Asian/Pacific Islander',
          'Hispanic',
          'Non-Hispanic Black',
          'Non-Hispanic White',
          'Other',
          'Unknown']

```

In this dataset, there are a large number of NaN values in the three columns below. The NaNs represent unknown or unreported information, not actual missing values in the sense of being zero. I also did not fill them with 0, since that would incorrectly imply there were no deaths.

In [15]: `print(df_copy.isna().sum())`

```

Year                0
Leading Cause        0
Sex                 0
Race/Ethnicity       0
Deaths             138
Death Rate          729
Age Adjusted Death Rate  729
dtype: int64

```

For data type conversion, I used `pd.to_numeric()` because it handles non-numeric mixed types, which can be an issue when the column was previously stored as an object type. Then I used `Int64` for the Deaths column so it does not fail the way a strict `int64` would, since `int64` requires fully clean numeric values. In contrast, `Int64` can handle NaN values while still preserving integer formatting.

In [16]: *### Data Type Conversion for Columns to Numeric: Deaths, Death Rate, Age Adjusted Death Rate*

```

df_copy[['Death Rate', 'Age Adjusted Death Rate']] = df_copy[['Death Rate', 'Age Adjusted Death Rate']]
df_copy['Deaths'] = df_copy['Deaths'].astype('Int64')

```

Now the dataset look much more cleaner!

```
In [17]: print(df_copy.head(-25))
```

	Year	Leading Cause	Sex \
0	2021	Mental and Behavioral Disorders due to Acciden...	Male
1	2021	All Other Causes	Female
2	2021	Chronic Liver Diseases and Cirrhosis	Male
3	2021	Diseases of Heart	Female
4	2021	Influenza	Male
...	...	...	...
2072	2007	Chronic Liver Disease and Cirrhosis	Male
2073	2007	Aortic Aneurysm and Dissection	Female
2074	2007	All Other Causes	Female
2075	2007	Diabetes Mellitus	Male
2076	2007	Alzheimer's Disease	Female

	Race/Ethnicity	Deaths	Death Rate	Age Adjusted Death Rate
0	Other	9	NaN	NaN
1	Unknown	102	NaN	NaN
2	Unknown	16	NaN	NaN
3	Hispanic	1409	111.0	95.5
4	Unknown	16	NaN	NaN
...	...	...	...	...
2072	Asian/Pacific Islander	18	3.7	3.6
2073	Unknown	<NA>	NaN	NaN
2074	Unknown	27	NaN	NaN
2075	Unknown	9	NaN	NaN
2076	Unknown	<NA>	NaN	NaN

```
[2077 rows x 7 columns]
```

I converted the data types into the correct numeric types:

```
In [18]: print(df_copy.dtypes)
```

```
Year                int64
Leading Cause       object
Sex                object
Race/Ethnicity     object
Deaths             Int64
Death Rate         float64
Age Adjusted Death Rate float64
dtype: object
```

```
In [19]: print(df_copy.describe())
```

	Year	Deaths	Death Rate	Age Adjusted Death Rate
count	2102.000000	1964.000000	1373.000000	1373.000000

mean	2014.089439	429.256110	56.258163	53.607305
std	4.333722	827.583725	75.598398	68.109884
min	2007.000000	1.000000	2.400000	2.500000
25%	2010.000000	25.000000	12.843000	12.338107
50%	2014.000000	140.000000	20.100000	19.800000
75%	2018.000000	317.250000	78.900000	81.500000
max	2021.000000	7050.000000	491.400000	414.594473

3 Analysis

Examining the age-adjusted death rates across race/ethnicity groups within leading causes:

In [20]: *### Top Causes by total deaths*

```
top_causes = (df_copy.groupby('Leading Cause')['Deaths']
               .sum()
               .sort_values(ascending = False)
               .head(5)
               .index)
```

Compare race/ethnicity within those causes

```
q1 = (df_copy[df_copy['Leading Cause'].isin(top_causes)]
      .groupby(['Leading Cause', 'Race/Ethnicity'])['Age Adjusted Death Rate'].mean())
```

Sort and Rank within each Cause

```
q1_sorted = q1.sort_values(['Leading Cause', 'Age Adjusted Death Rate'], ascending =
                             False)
print(q1_sorted)
```

	Leading Cause	Race/Ethnicity	Age Adjusted Death Rate
5	All Other Causes	Unknown	163.585872
2	All Other Causes	Non-Hispanic Black	153.113634
1	All Other Causes	Hispanic	119.606148
3	All Other Causes	Non-Hispanic White	118.451557
0	All Other Causes	Asian/Pacific Islander	69.627125
4	All Other Causes	Other	NaN
7	Covid-19	Hispanic	190.475000
8	Covid-19	Non-Hispanic Black	184.925000
6	Covid-19	Asian/Pacific Islander	110.500000
9	Covid-19	Non-Hispanic White	100.400000
10	Covid-19	Other	NaN
11	Covid-19	Unknown	NaN
17	Diseases of Heart	Unknown	299.247153
14	Diseases of Heart	Non-Hispanic Black	240.924422
15	Diseases of Heart	Non-Hispanic White	216.471115

13	Diseases of Heart	Hispanic	156.998835
12	Diseases of Heart	Asian/Pacific Islander	109.716362
16	Diseases of Heart	Other	NaN
20	Influenza	Non-Hispanic Black	27.112582
21	Influenza	Non-Hispanic White	23.846512
19	Influenza	Hispanic	23.588610
23	Influenza	Unknown	22.709890
18	Influenza	Asian/Pacific Islander	18.850101
22	Influenza	Other	NaN
26	Malignant Neoplasms	Non-Hispanic Black	171.326031
27	Malignant Neoplasms	Non-Hispanic White	166.598800
29	Malignant Neoplasms	Unknown	131.043410
25	Malignant Neoplasms	Hispanic	118.639732
24	Malignant Neoplasms	Asian/Pacific Islander	101.710984
28	Malignant Neoplasms	Other	NaN

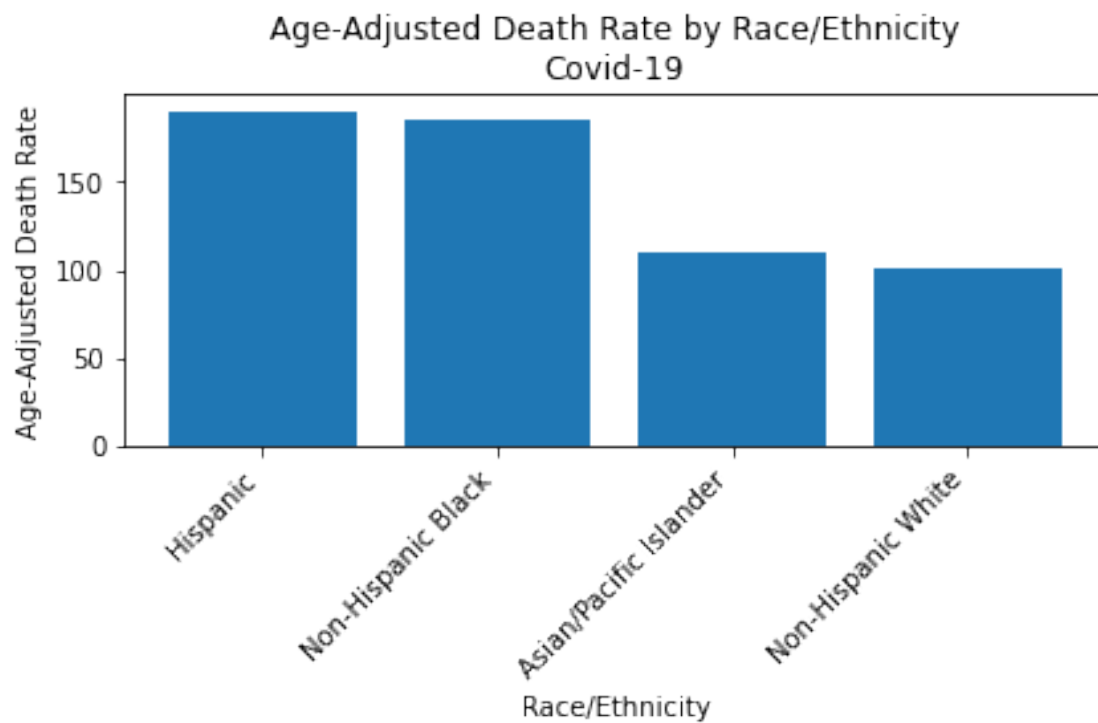
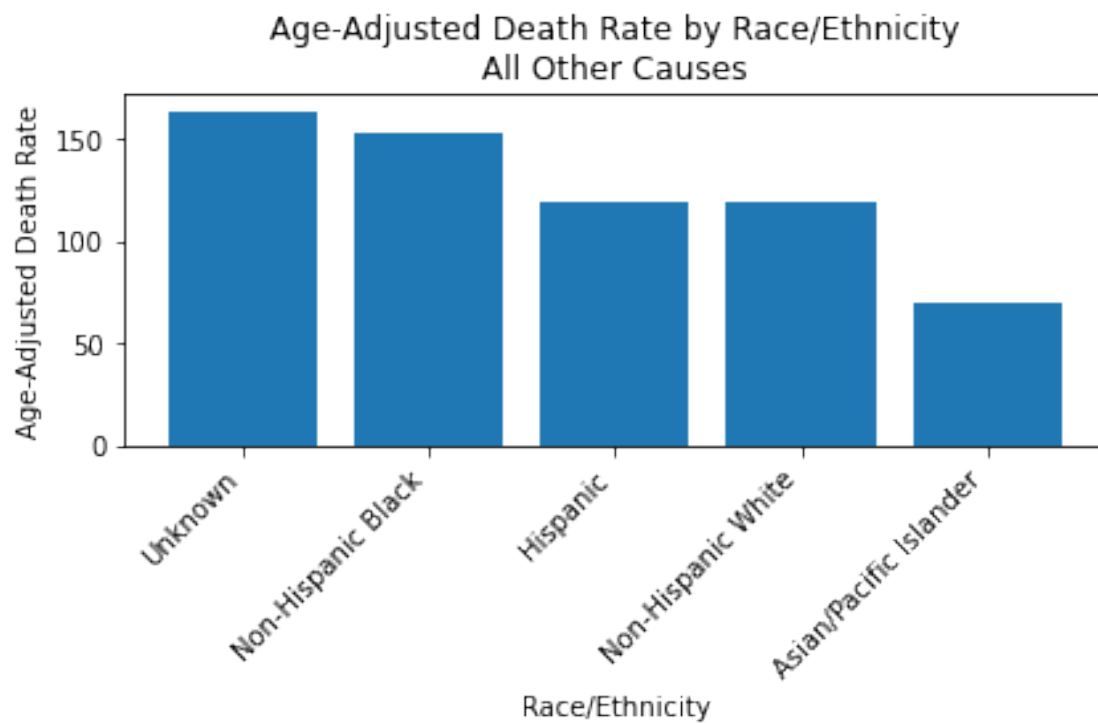
```
In [21]: top_causes = q1_sorted['Leading Cause'].unique()

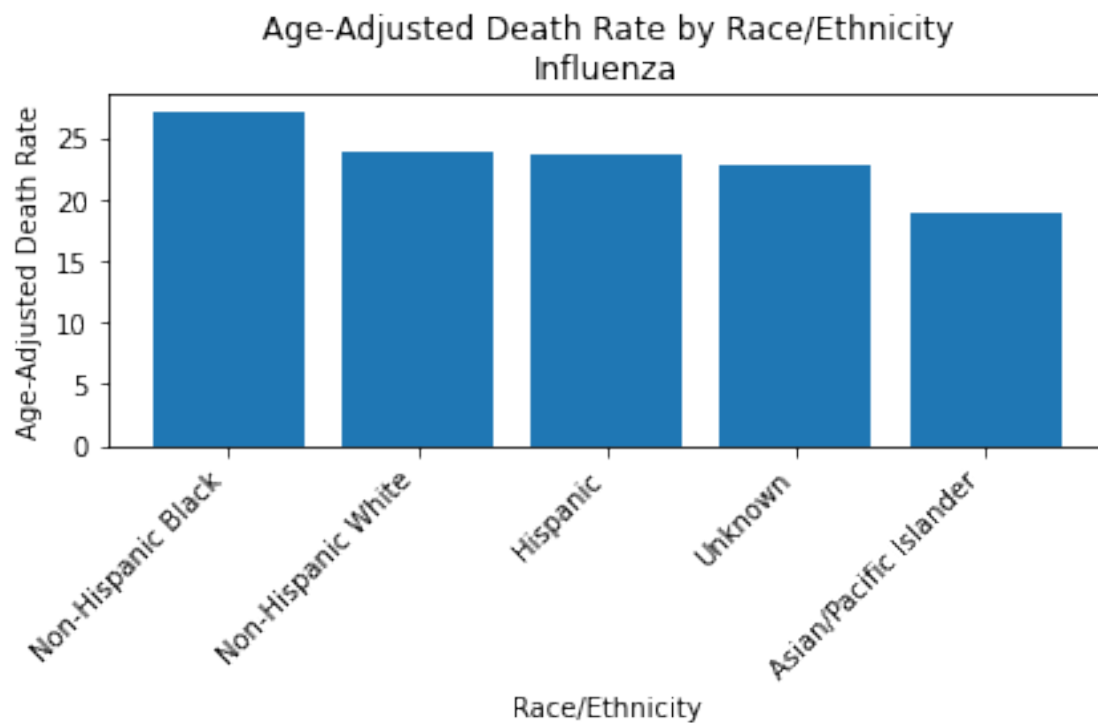
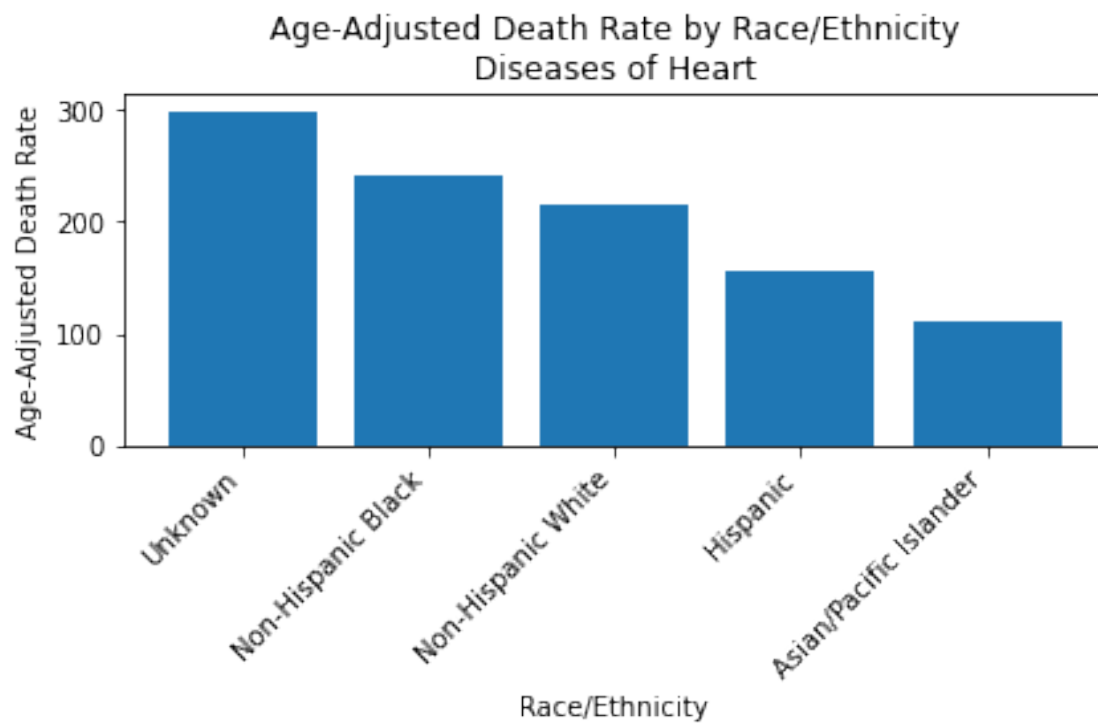
for cause in top_causes:
    temp = q1_sorted[q1_sorted['Leading Cause'] == cause]

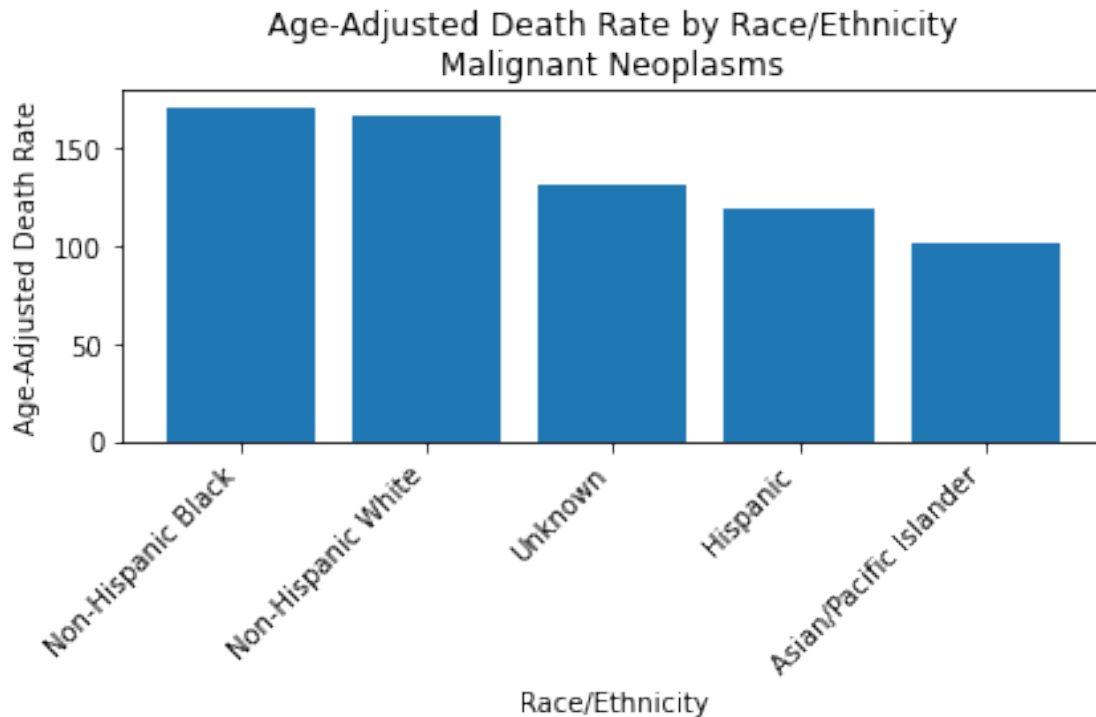
    plt.figure()
    plt.bar(temp['Race/Ethnicity'], temp['Age Adjusted Death Rate'])

    plt.title(f'Age-Adjusted Death Rate by Race/Ethnicity\n{n}{cause}')
    plt.xlabel('Race/Ethnicity')
    plt.ylabel('Age-Adjusted Death Rate')

    plt.xticks(rotation=45, ha='right')
    plt.tight_layout()
    plt.show()
```







Displayed above, we can see that the highest at-risk demographic group across several leading causes of death such as Heart Disease, Malignant Neoplasms, All Other Causes, and Influenza is the Non-Hispanic Black population. The exception is COVID-19, where Hispanic individuals show the highest age-adjusted death rates, with Non-Hispanic Black individuals still remaining among the most affected groups in New York City.

According to the standard epidemiological scaling convention (CDC: <https://www.cdc.gov/nchs/has/sources-definitions/rate.htm>), age-adjusted death rates are conveyed as mortality per 100,000 population. This means that the value 240.924422 shown in the output and visualizations for diseases of the heart among Non-Hispanic Black individuals corresponds to about 241 deaths per 100,000 people, after adjusting for differences in age distribution across groups. This makes the comparisons between demographic groups more fair and meaningful.

I also included the Unknown category in both the visuals and the code to show that a noticeable amount of race/ethnicity data was not reported or recorded. This missing information could potentially introduce some bias into the results, since excluding or misclassifying these cases may affect the overall distribution of leading causes of death across different demographic groups.

Differences in Age-Adjusted Death Rates Between Males and Females Within the Highest-Risk Demographic Groups:

In [22]: *### Create top leading causes for affected groups*

```
top = q1_sorted.groupby('Leading Cause').head(1)
top_groups = top[['Leading Cause', 'Race/Ethnicity']]
```

```
### Filter Original Data
```

```
df_q2 = df_copy.merge(top_groups, on = ['Leading Cause', 'Race/Ethnicity'])
```

```
### Compare By Sex
```

```
q2 = (df_q2.groupby(['Leading Cause', 'Race/Ethnicity', 'Sex'])['Age Adjusted Death Rate']
      .mean()
      .unstack())
```

```
### Gender Gap
```

```
q2['Gap (Male - Female)'] = q2['Male'] - q2['Female']
```

```
print(q2)
```

Sex		Female	Male \
Leading Cause	Race/Ethnicity		
All Other Causes	Unknown	94.641463	232.530281
Covid-19	Hispanic	121.500000	259.450000
Diseases of Heart	Unknown	194.315070	404.179237
Influenza	Non-Hispanic Black	21.244228	32.980936
Malignant Neoplasms	Non-Hispanic Black	145.144305	197.507757

Sex		Gap (Male - Female)
Leading Cause	Race/Ethnicity	
All Other Causes	Unknown	137.888818
Covid-19	Hispanic	137.950000
Diseases of Heart	Unknown	209.864167
Influenza	Non-Hispanic Black	11.736708
Malignant Neoplasms	Non-Hispanic Black	52.363452

```
In [23]: q2_plot = q2.reset_index()
```

```
plt.figure(figsize=(12, 6))
```

```
x = range(len(q2_plot))
```

```
width = 0.35
```

```
plt.bar(x, q2_plot['Male'], width=width, label='Male')
```

```
plt.bar([i + width for i in x], q2_plot['Female'], width=width, label='Female')
```

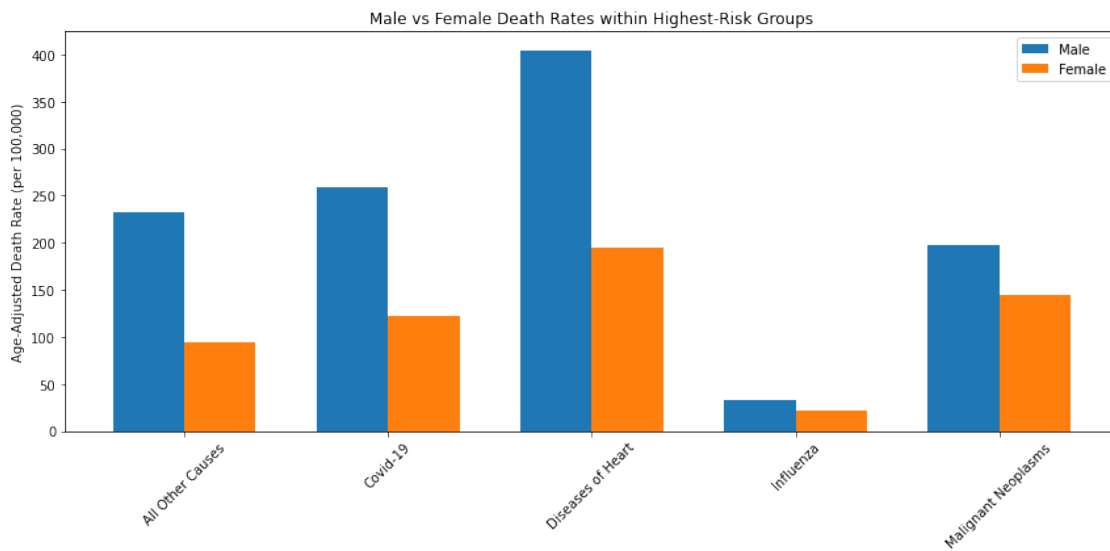
```
plt.xticks([i + width/2 for i in x], q2_plot['Leading Cause'], rotation=45)
```

```
plt.ylabel('Age-Adjusted Death Rate (per 100,000)')
```

```
plt.title('Male vs Female Death Rates within Highest-Risk Groups')
```

```
plt.legend()
```

```
plt.tight_layout()
plt.show()
```



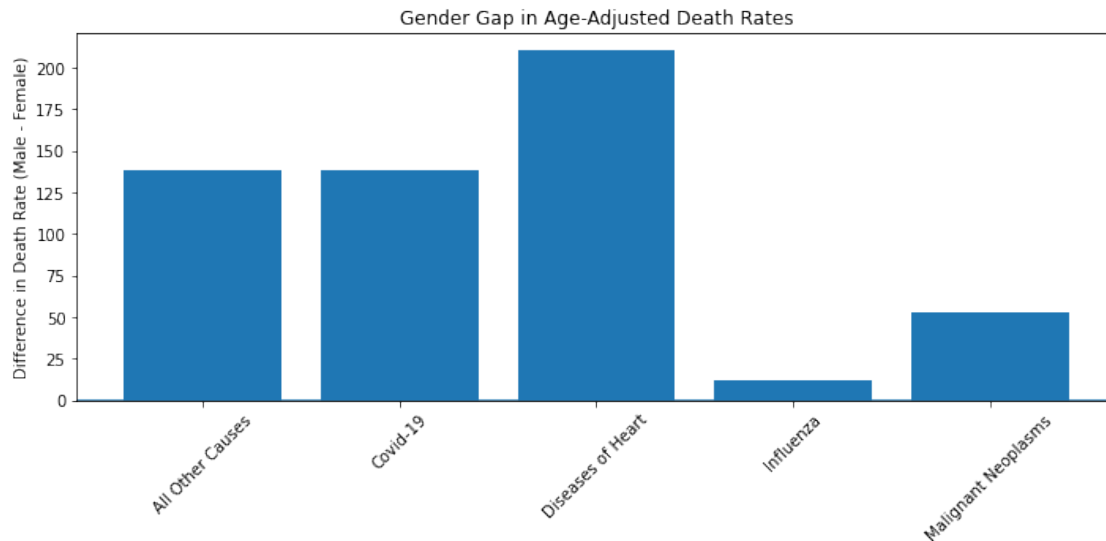
```
In [24]: plt.figure(figsize=(10, 5))

plt.bar(q2_plot['Leading Cause'], q2_plot['Gap (Male - Female)'])

plt.axhline(0) # baseline

plt.xticks(rotation=45)
plt.ylabel('Difference in Death Rate (Male - Female)')
plt.title('Gender Gap in Age-Adjusted Death Rates')

plt.tight_layout()
plt.show()
```



Looking at the results and comparing genders across leading causes of death, men consistently have higher mortality risk in all categories. The largest difference appears in heart disease, where the age-adjusted death rate for men is about 404 per 100,000 people, compared to 194 per 100,000 for women - nearly twice as high. This further highlights the disparity between genders, as shown in the “Gender Gap in Age-Adjusted Death Rates” chart, where all values are positive. This indicates that males have higher death rates across every category, with no instances where female rates exceed those of males.

Additionally, the observed gaps show that, for causes such as All Other Causes, COVID-19, and Diseases of the Heart, female death rates do not reach even half of the male rates. While Malignant Neoplasms and Influenza show smaller differences, the overall pattern of higher mortality among men remains consistent. These results suggest a persistent gender disparity in health outcomes across multiple leading causes of death.

Age-adjusted Death Rates between Race/Ethnicity Groups over Time For Top Leading Causes:

In [25]: *### Filter to top causes*

```
top_causes3 = q1_sorted['Leading Cause'].unique()
df_q3 = df_copy[df_copy['Leading Cause'].isin(top_causes3)]
```

Group by time, cause, and race

```
q3 = (df_q3.groupby(['Year', 'Leading Cause', 'Race/Ethnicity'])['Age Adjusted Death Rate']
      .mean()
      .reset_index())
```

Measure disparity

```
disparity = (q3.groupby(['Year', 'Leading Cause'])['Age Adjusted Death Rate']
             .agg(lambda x: x.max() - x.min())
             .reset_index(name = 'Disparity'))
```

```
pd.set_option('display.max_rows', None)
```

```
print(disparity)
```

	Year	Leading Cause	Disparity
0	2007	All Other Causes	85.000000
1	2007	Diseases of Heart	164.200000
2	2007	Influenza	12.750000
3	2007	Malignant Neoplasms	78.400000
4	2008	All Other Causes	83.000000
5	2008	Diseases of Heart	151.900000
6	2008	Influenza	8.450000
7	2008	Malignant Neoplasms	89.700000
8	2009	All Other Causes	90.000000
9	2009	Diseases of Heart	139.250000
10	2009	Influenza	5.700000
11	2009	Malignant Neoplasms	84.700000
12	2010	All Other Causes	77.400000
13	2010	Diseases of Heart	141.100000
14	2010	Influenza	6.900000
15	2010	Malignant Neoplasms	87.600000
16	2011	All Other Causes	84.000000
17	2011	Diseases of Heart	125.550000
18	2011	Influenza	9.100000
19	2011	Malignant Neoplasms	85.600000
20	2012	All Other Causes	78.800000
21	2012	Diseases of Heart	133.250000
22	2012	Influenza	12.750000
23	2012	Malignant Neoplasms	76.100000
24	2013	All Other Causes	78.200000
25	2013	Diseases of Heart	120.150000
26	2013	Influenza	10.550000
27	2013	Malignant Neoplasms	73.900000
28	2014	All Other Causes	77.400000
29	2014	Diseases of Heart	117.100000
30	2014	Influenza	7.700000
31	2014	Malignant Neoplasms	68.950000
32	2015	All Other Causes	82.523811
33	2015	Diseases of Heart	119.548795
34	2015	Influenza	6.452239
35	2015	Malignant Neoplasms	70.080536
36	2016	All Other Causes	76.162290
37	2016	Diseases of Heart	120.911594
38	2016	Influenza	6.569921
39	2016	Malignant Neoplasms	70.267077
40	2017	All Other Causes	72.338281
41	2017	Diseases of Heart	120.442976

42	2017	Influenza	10.747745
43	2017	Malignant Neoplasms	53.423314
44	2018	All Other Causes	89.448500
45	2018	Diseases of Heart	209.302000
46	2018	Influenza	9.440500
47	2018	Malignant Neoplasms	62.025500
48	2019	All Other Causes	98.017235
49	2019	Diseases of Heart	189.020475
50	2019	Influenza	9.126814
51	2019	Malignant Neoplasms	59.579285
52	2020	All Other Causes	99.850000
53	2020	Covid-19	152.500000
54	2020	Diseases of Heart	157.500000
55	2020	Influenza	6.400000
56	2020	Malignant Neoplasms	43.550000
57	2021	All Other Causes	99.450000
58	2021	Covid-19	27.650000
59	2021	Diseases of Heart	118.650000
60	2021	Influenza	6.950000
61	2021	Malignant Neoplasms	42.200000

The table above shows the highest and lowest leading causes of death by race/ethnicity for each year from 2007 to 2021 based on age-adjusted death rates. Additionally, each value in the “Disparity” column measures the differences in death rates across race/ethnicity for a given year and cause of death.

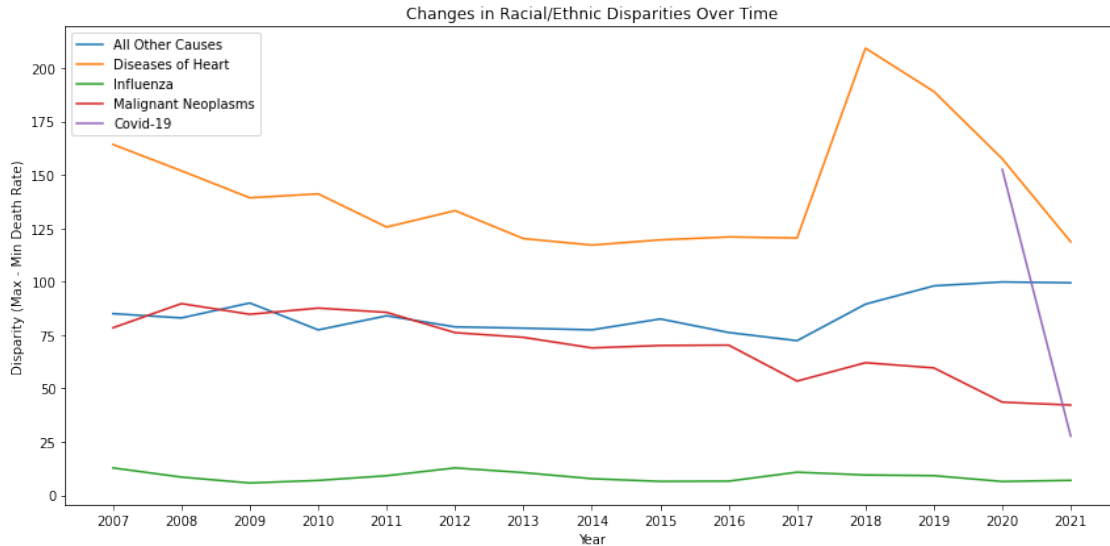
```
In [26]: plt.figure(figsize=(12, 6))

for cause in disparity['Leading Cause'].unique():
    subset = disparity[disparity['Leading Cause'] == cause]
    plt.plot(subset['Year'], subset['Disparity'], label=cause)

plt.xlabel('Year')
plt.ylabel('Disparity (Max - Min Death Rate)')
plt.title('Changes in Racial/Ethnic Disparities Over Time')

plt.xticks(range(2007, 2022))

plt.legend()
plt.tight_layout()
plt.show()
```



In this line chart, Diseases of Heart remains the number 1 cause of death disparity consistently over time. From 2017 to 2018, there is a drastic upward trend in heart disease, increasing from about 120.44 to 209.30 per 100,000 individuals in New York City. Then from 2018 to 2019, after reaching the peak of 209.30, the disparity decreases slightly to about 189.02. These two years stand out as an outlier period, showing the largest magnitude change compared to all other causes.

In the earlier period from 2007 to 2014, there is a steady decline from about 164.20 down to 117.10, suggesting some improvement in inequality over time. From 2015 to 2017, the disparity levels for heart disease stay relatively stable, hovering around the 118–121 range. In 2020, the trend increases again to about 157.50, before dropping in 2021 to around 118.65, which is close to earlier baseline levels seen in the mid-2010's.

When comparing other leading causes like All Other Causes and Malignant Neoplasms, their values stay relatively closer in range and much more stable overall, with smaller fluctuations and gradual changes over time. Malignant Neoplasms generally shows a slight downward trend in the later years, while All Other Causes remains fairly steady with moderate variation.

COVID-19 in 2020 is a major exception, where its disparity is close to Heart Disease (152.50 vs. 157.50). However, in 2021, COVID-19 drops significantly to about 27.65, showing a sharp decrease across demographic groups.

Lastly, Influenza does not show any significant disparity patterns, with values staying consistently low and flat throughout the years, roughly in the 5–12 range.

4 Conclusion

Key Findings:

1. Heart disease is the leading cause of death disparity across all demographic groups.
2. The most at-risk ethnic group is Non-Hispanic Black across the top five causes, with the exception of COVID-19, where rates are comparable to the Hispanic population.
3. Males disproportionately exceed females in the leading causes of death, showing a noticeable disparity gap that is often close to twice the magnitude.

4. Diseases of Heart consistently has the highest mortality disparity rate year over year from 2007 to 2021.

Limitations:

The dataset appears to be missing important details that could provide a more accurate representation of population health and leading causes of death. In particular, factors such as socioeconomic status are not included, even though they likely play a significant role in mortality disparities.

Additionally, the category “All Other Causes” is too broad and vague, which limits deeper analysis. Grouping multiple causes together makes it difficult to identify specific contributors to mortality across different racial and ethnic groups. This aggregation may also distort interpretation by understating the true significance of these underlying causes.